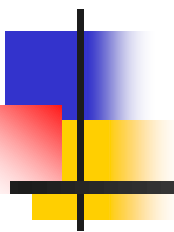


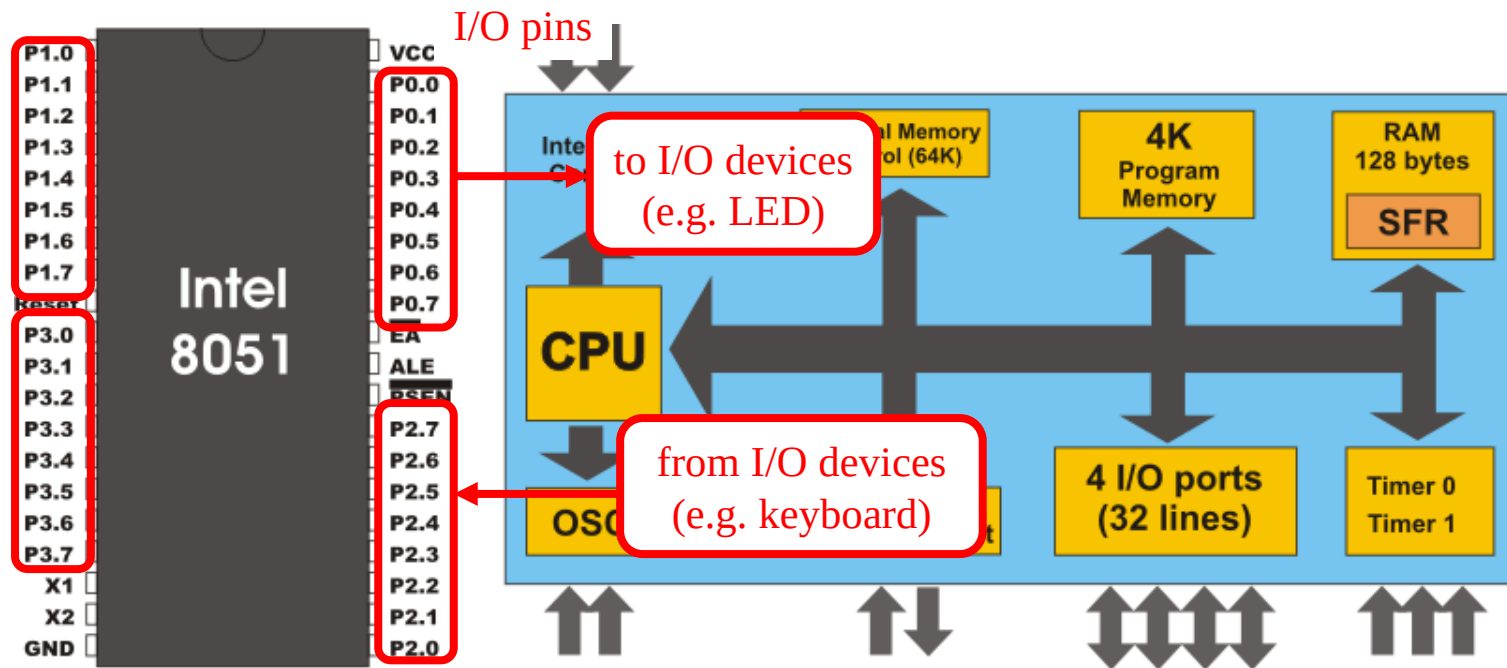
Lab 02

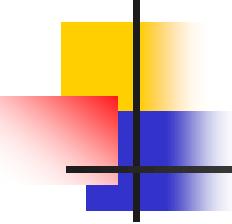
General Purpose Digital I/O (GPIO)



Objectives of this lab

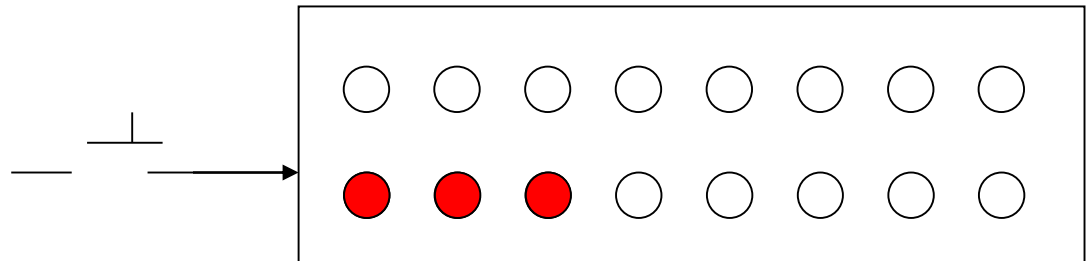
- (1) To build up your imagination on how a program affects hardware signals
- (2) To learn how to send/receive signals from an application processor to external devices through I/O pads

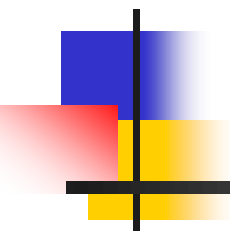




Your work

- design a LED box
 - initial: all LED off
 - the LED runs some pattern after some button pressed
 - you can design your own pattern

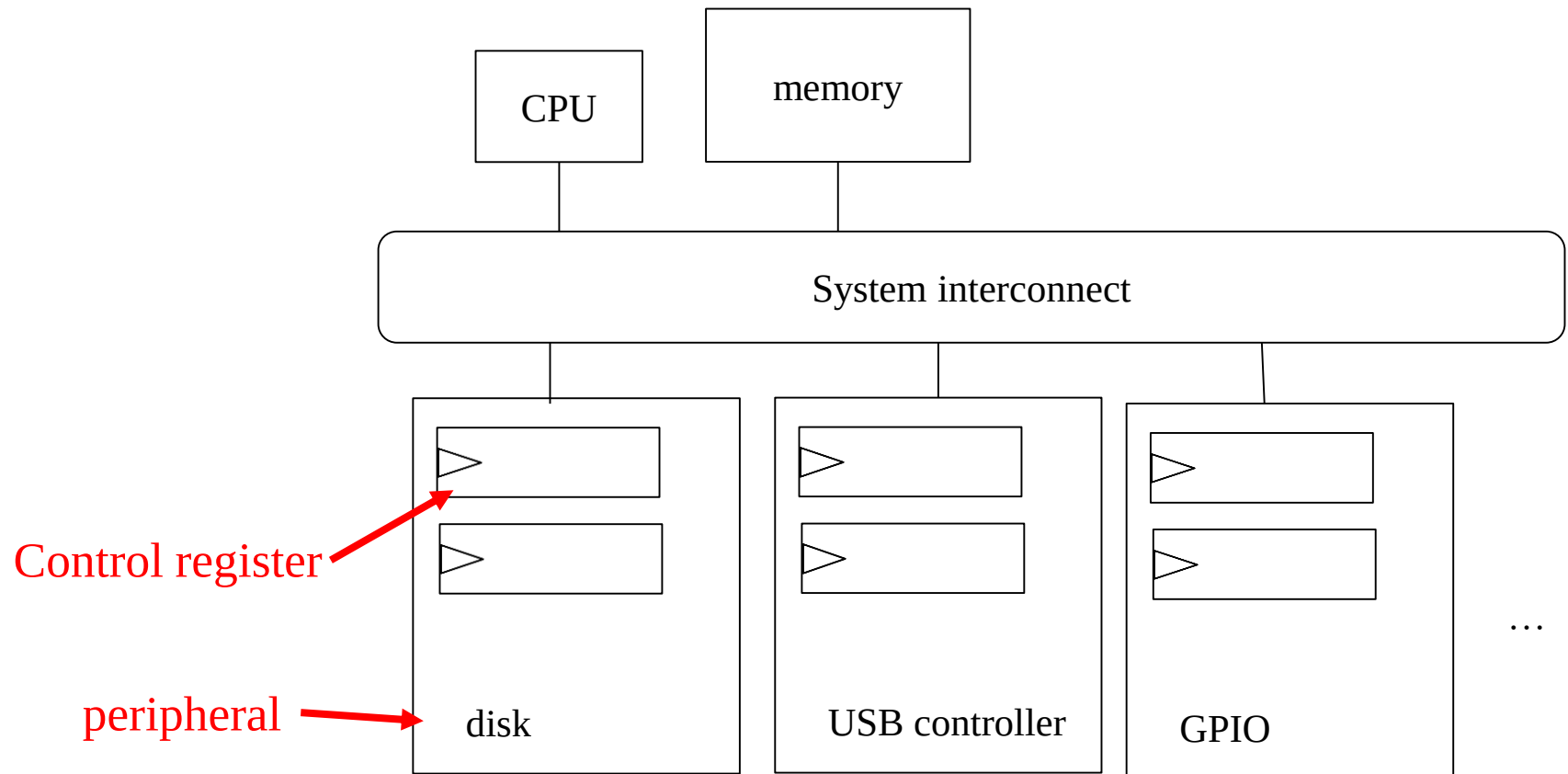




General I/O Control Model

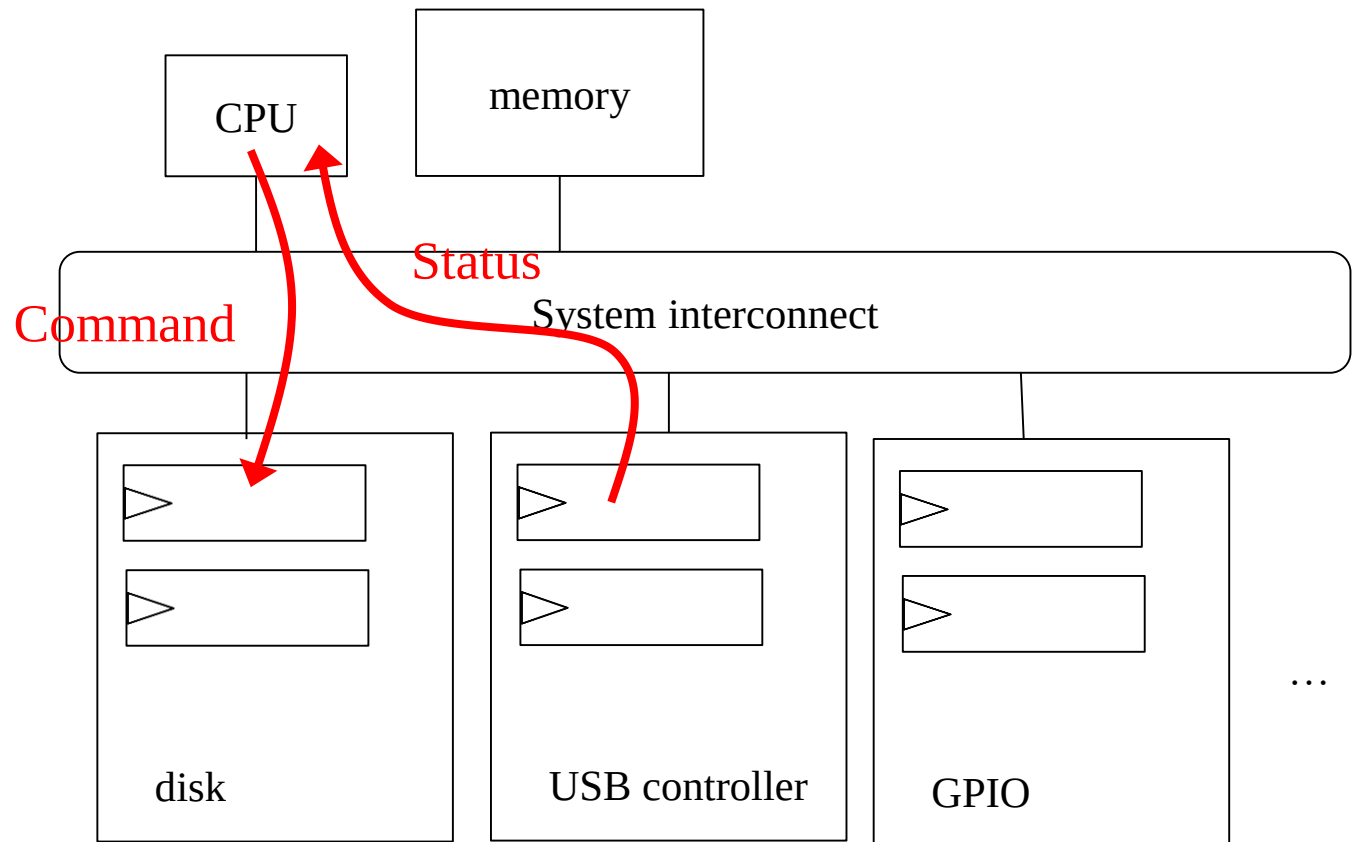
How a processor commands an I/O peripheral

- Through access control registers



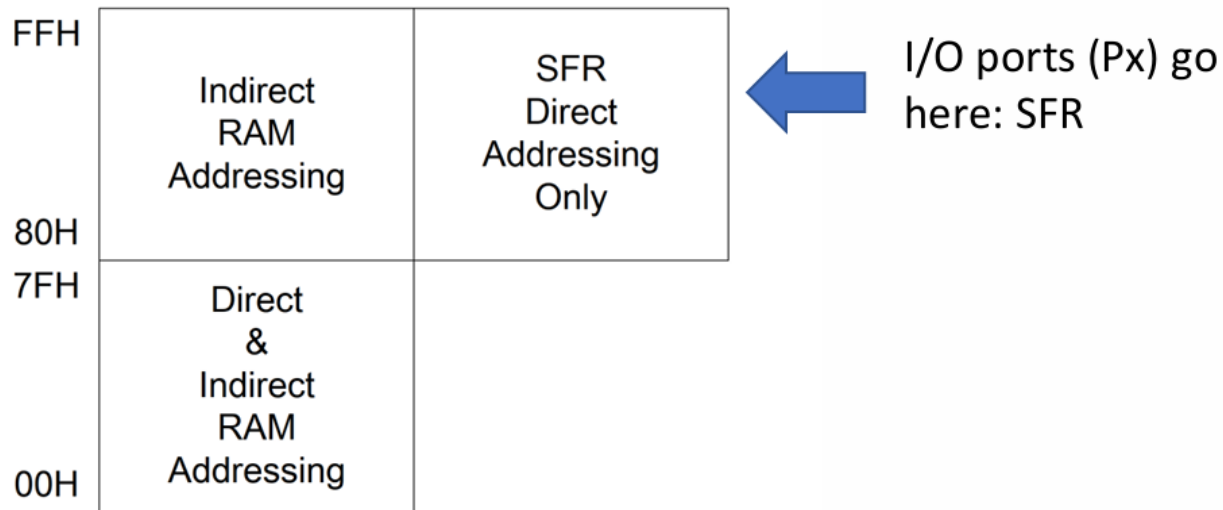
How a processor commands an I/O peripheral

- Through access control registers



How to access control registers: the memory-mapped I/O

- Part of the addressing space is assigned to control registers
- Each control register is mapped to some memory address

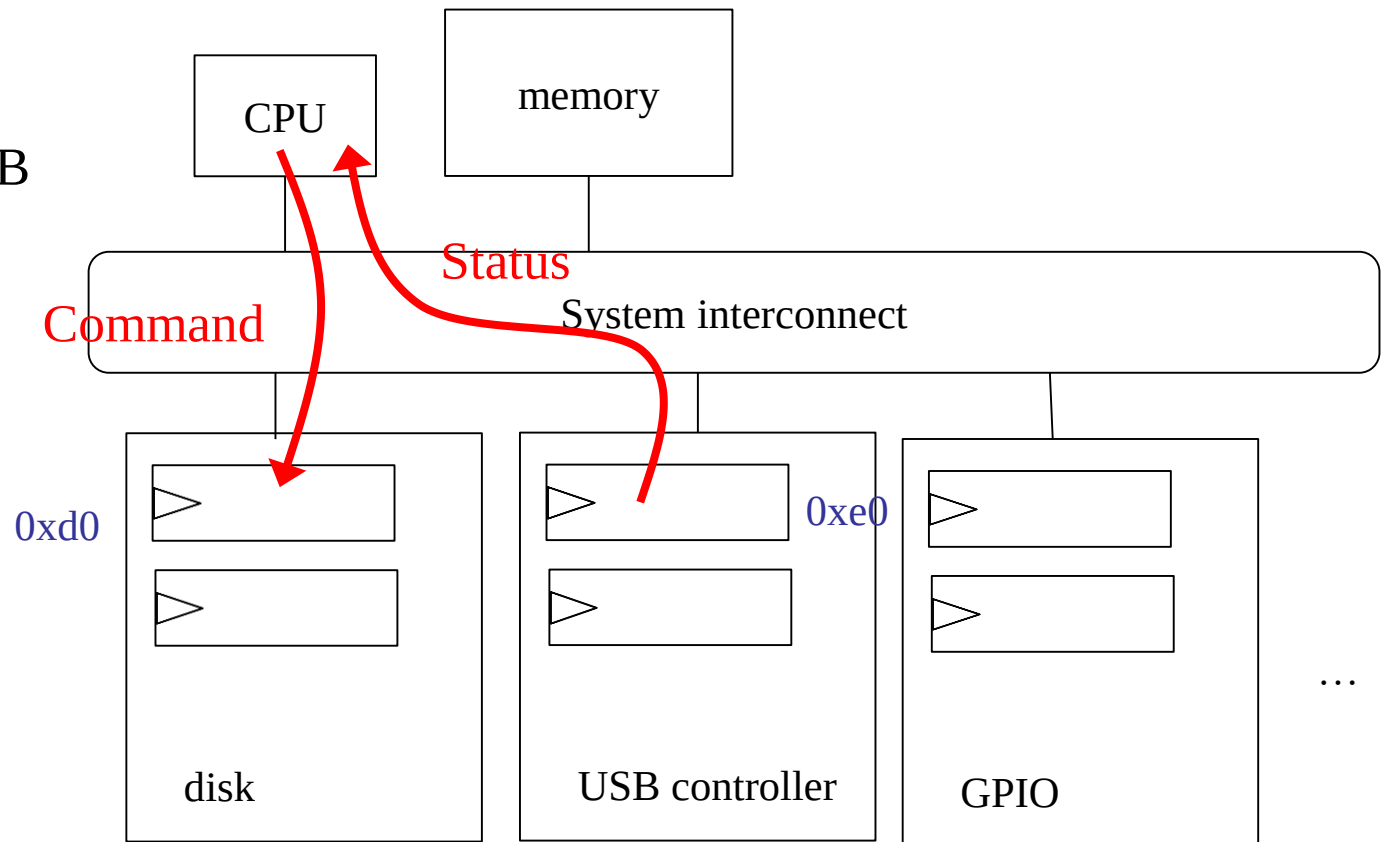


256 bytes RAM and SFR Data Memory Space

How a processor commands an I/O peripheral

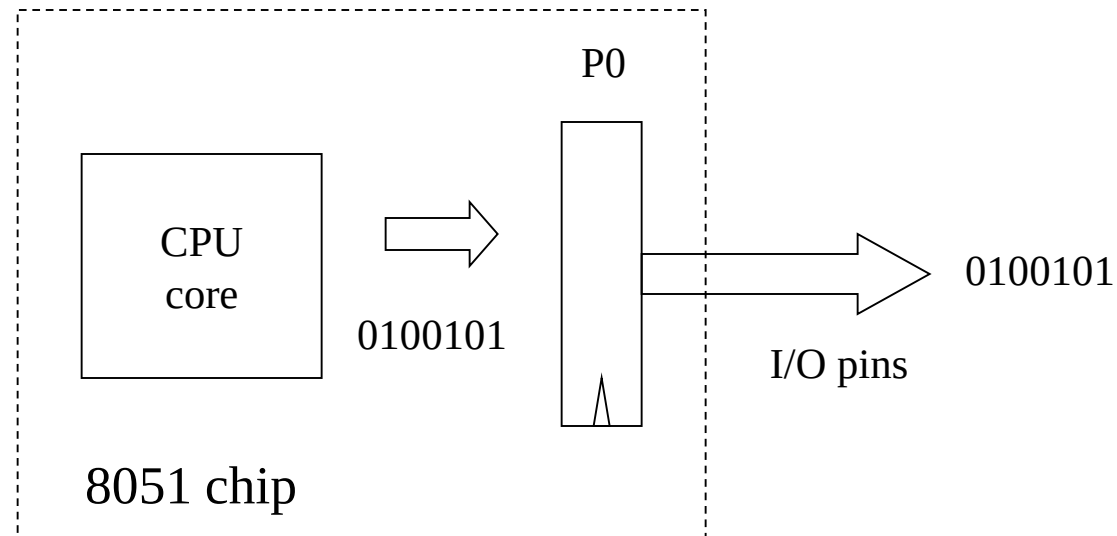
- Through access control registers

```
mov 0xd0, #0101B  
mov R1, 0xe0
```

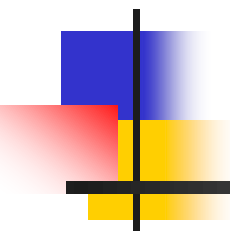


General Purpose Digital I/O

- The processor assigns/examines the logical status of some I/O pins directly

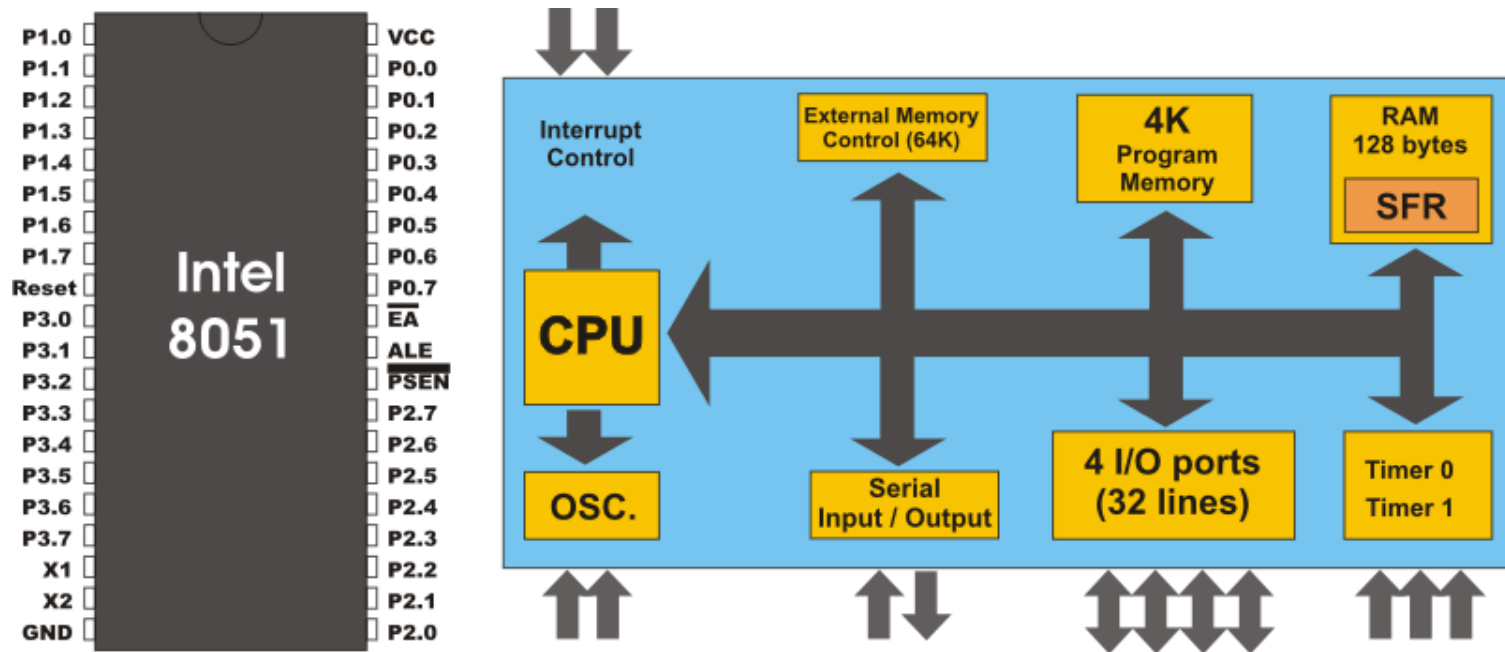


I/O Model of Legacy 8051 Processor



Features of 8051 I/O

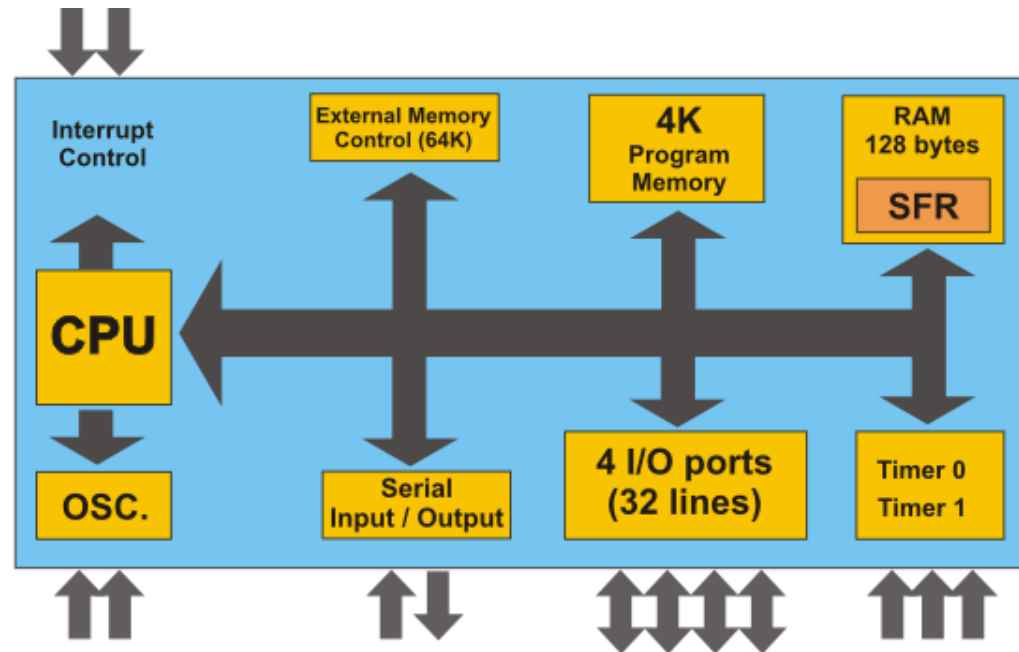
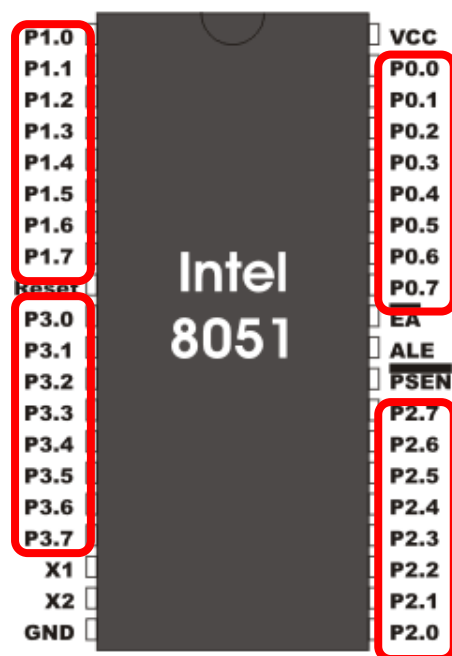
- (1) Four 8-bit I/O ports P0-P3
- (2) Each pin is bidirectional
 - sometimes input and sometimes output



Features of 8051 I/O

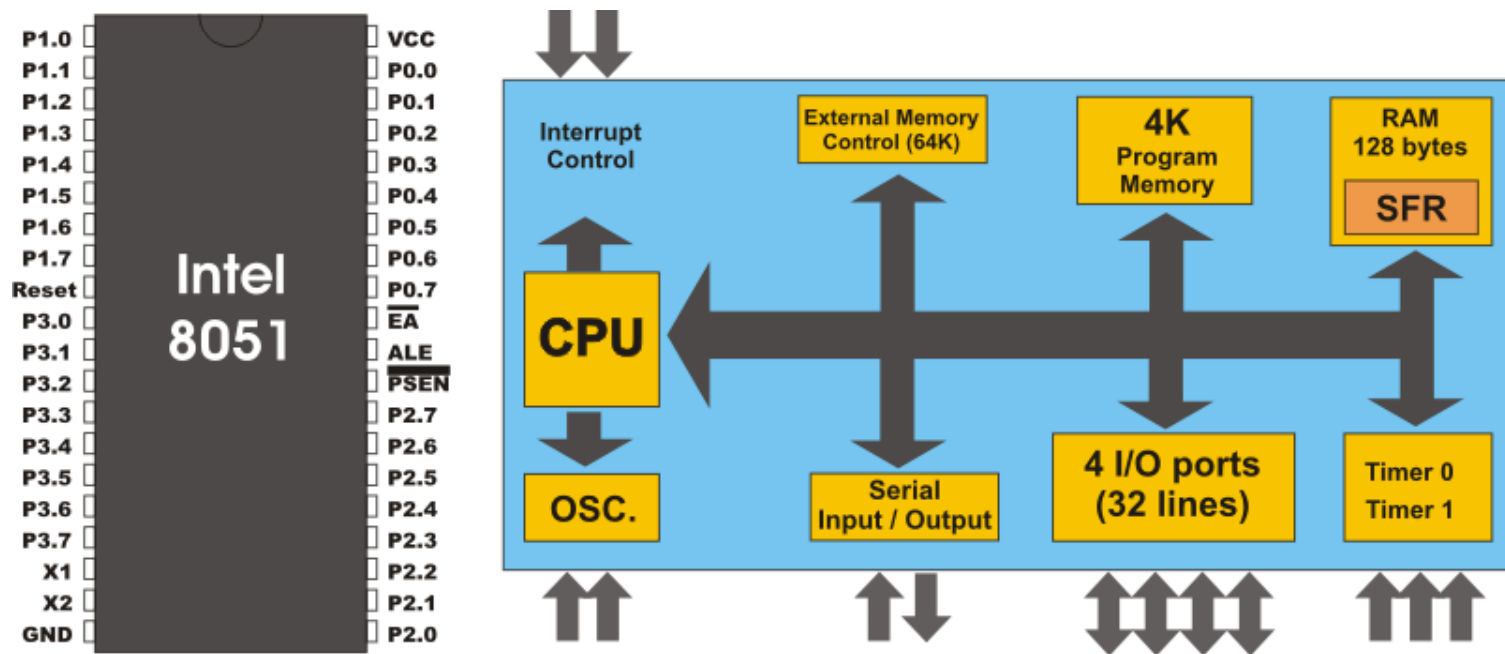
- (1) Four 8-bit I/O ports P0-P3
- (2) Each pin is bidirectional
 - sometimes input and sometimes output

Note: in our 8052-like architecture, an additional P4 can be used. However, the basic principle is the same.



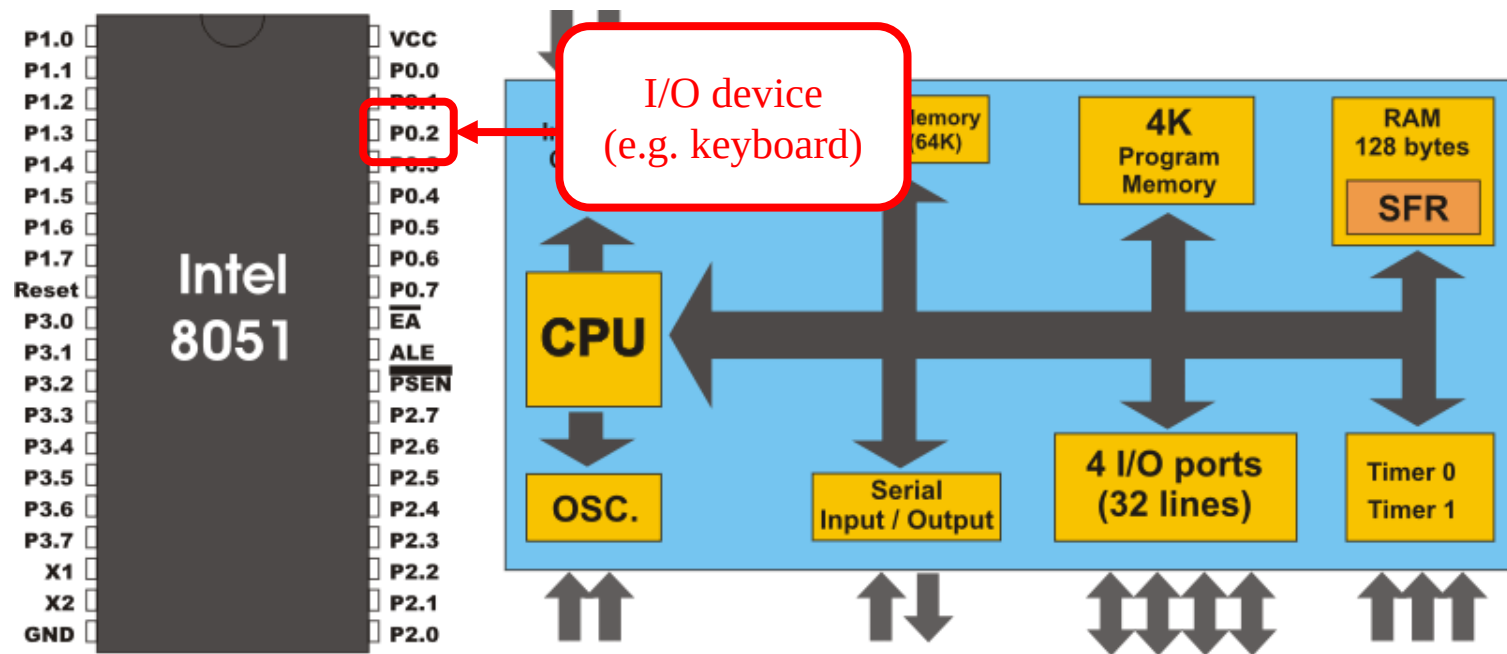
Features of 8051 I/O

- (1) Four 8-bit I/O ports P0-P3
- (2) **Each pin is bidirectional**
 - sometimes input and sometimes output



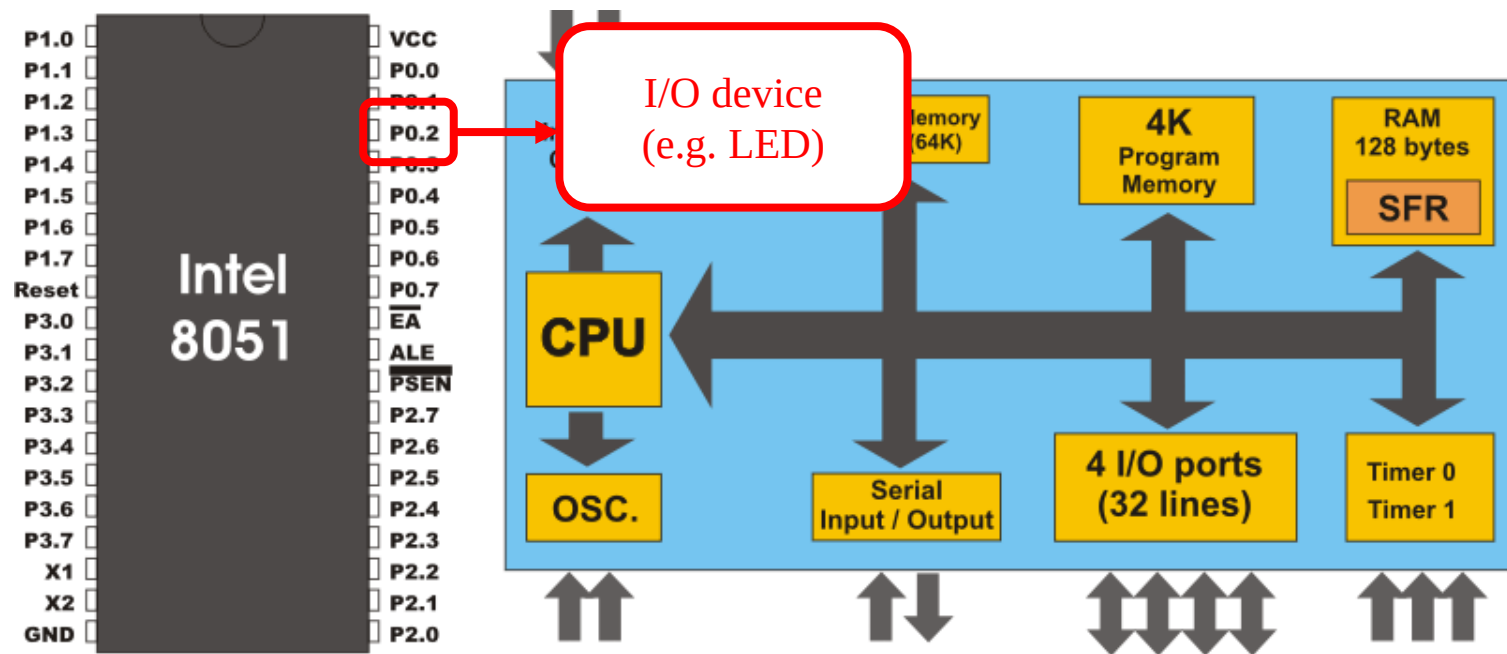
Features of 8051 I/O

- (1) Four 8-bit I/O ports P0-P3
- (2) Each pin is bidirectional
 - Sometimes input and sometimes output



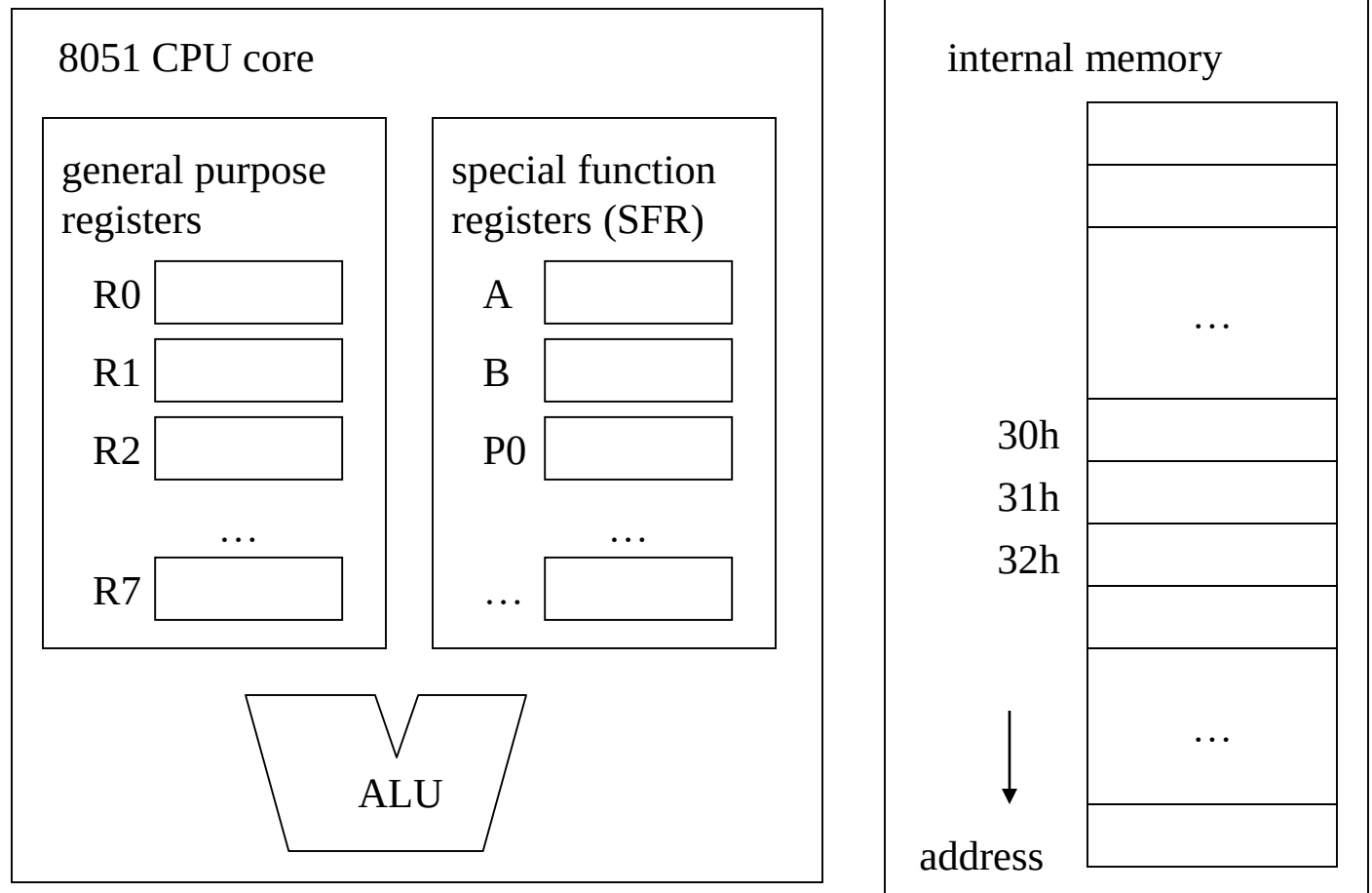
Features of 8051 I/O

- (1) Four 8-bit I/O ports P0-P3
- (2) Each pin is bidirectional
 - sometimes input and sometimes output



Imagination on 8051 architecture

- Imagine how data flow in the architecture!



How to program I/O ports?

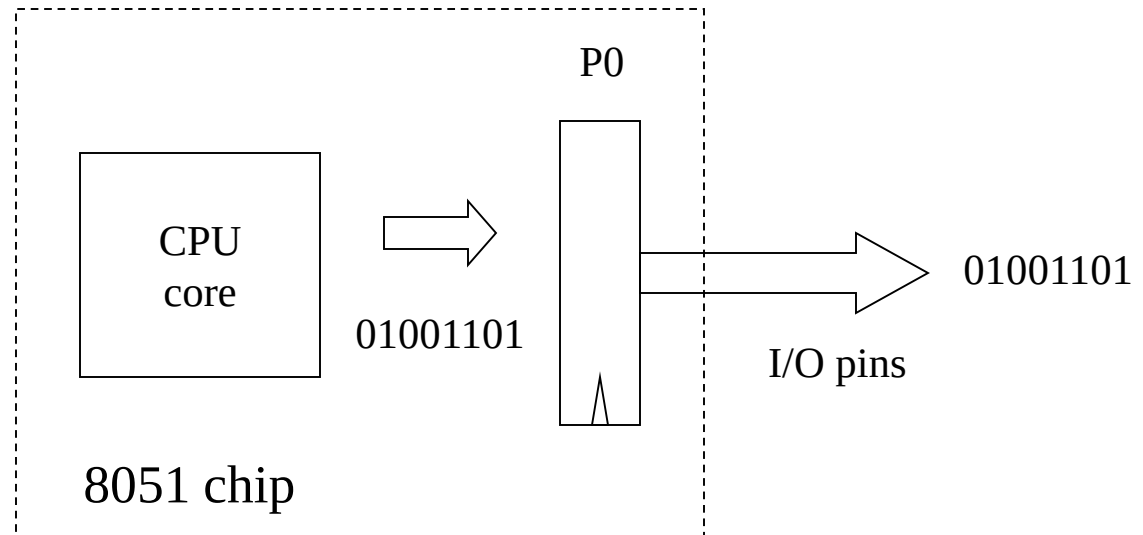
- through SFRs P0-P3

F8									FF
F0	B								F7
E8									EF
E0	ACC								E7
D8									DF
D0	PSW								D7
C8									CF
C0									C7
B8	IP								BF
B0	P3								B7
A8	IE								AF
A0	P2								A7
98	SCON	SBUF							9F
90	P1								97
88	TCON	TMOD	TL0	TL1	TH0	TH1			8F
80	P0	SP	DPL	DPH				PCON	87

Bit-addressable Registers

How 8051 send out dedicated control signals

```
MOV R0, #01001101B  
MOV P0, R0
```



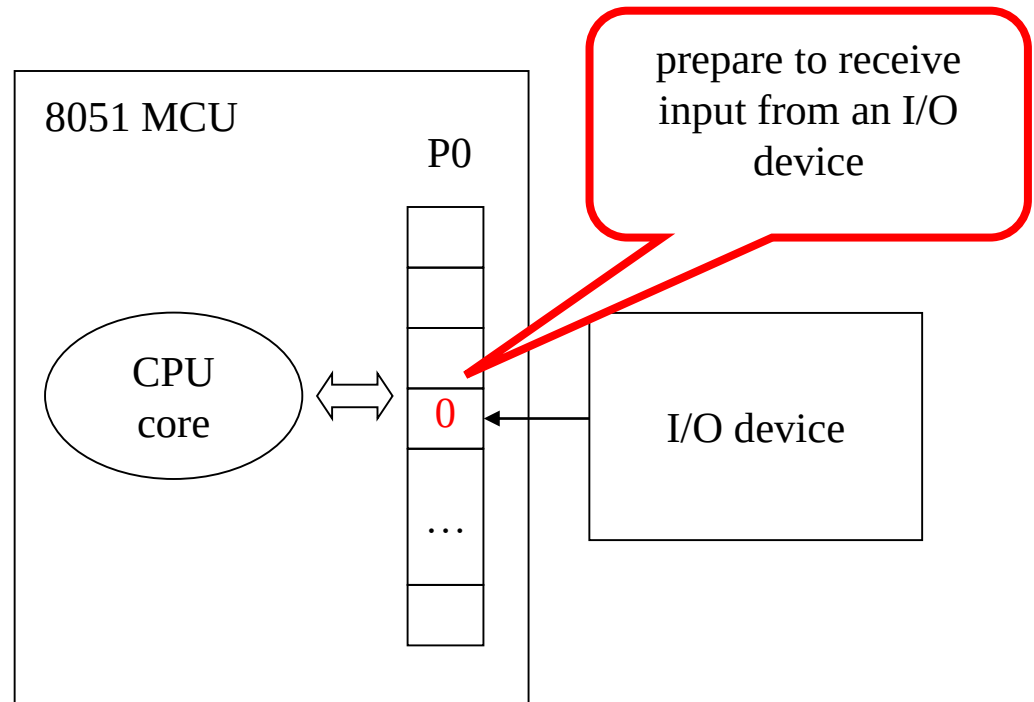
The case of input (receive)

- initial: set a bit (pin) with value 0
- receive (input): wait for the bit to be toggled to be 1

P0.3 = 0

```
//wait unit P0.3 been set to 1  
while (P0.3==0);
```

```
//action for the I/O event
```



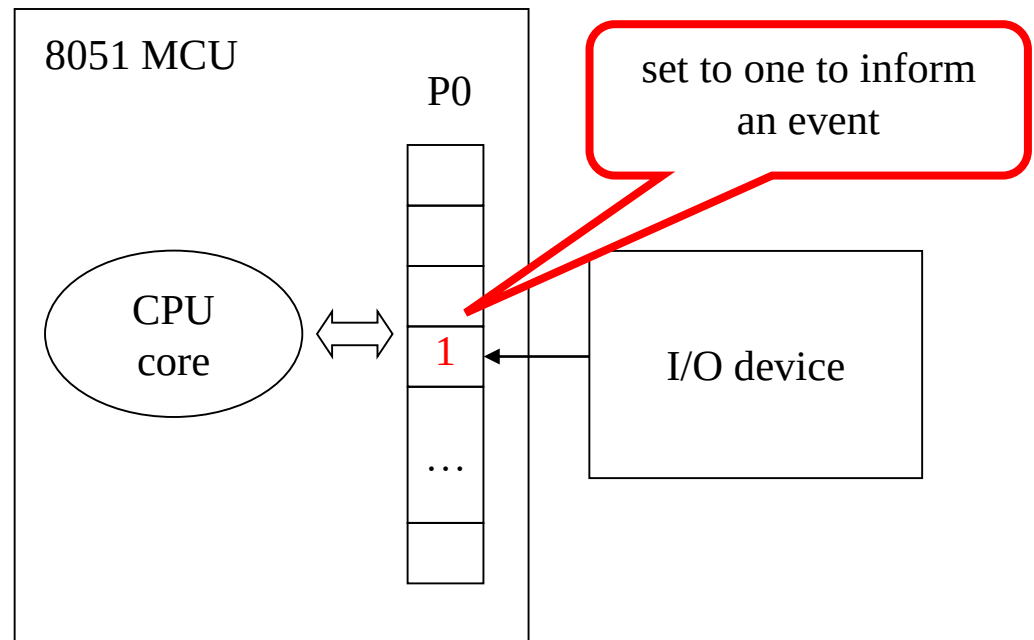
The case of input (receive)

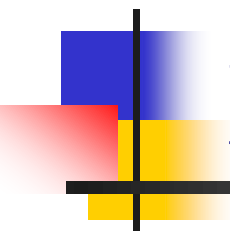
- initial: set a bit (pin) with value 0
- receive (input): wait for the bit to be toggled to be 1

P0.3 = 0

```
//wait unit P0.3 been set to 1  
while (P0.3==0);
```

```
//action for the I/O event
```





Example 1: wait for a button pressed

Show how to input signal

Demo: wait for a button pressed

wait:

A = P1;

if (A==0) goto wait;

exit:

//something after button pressed

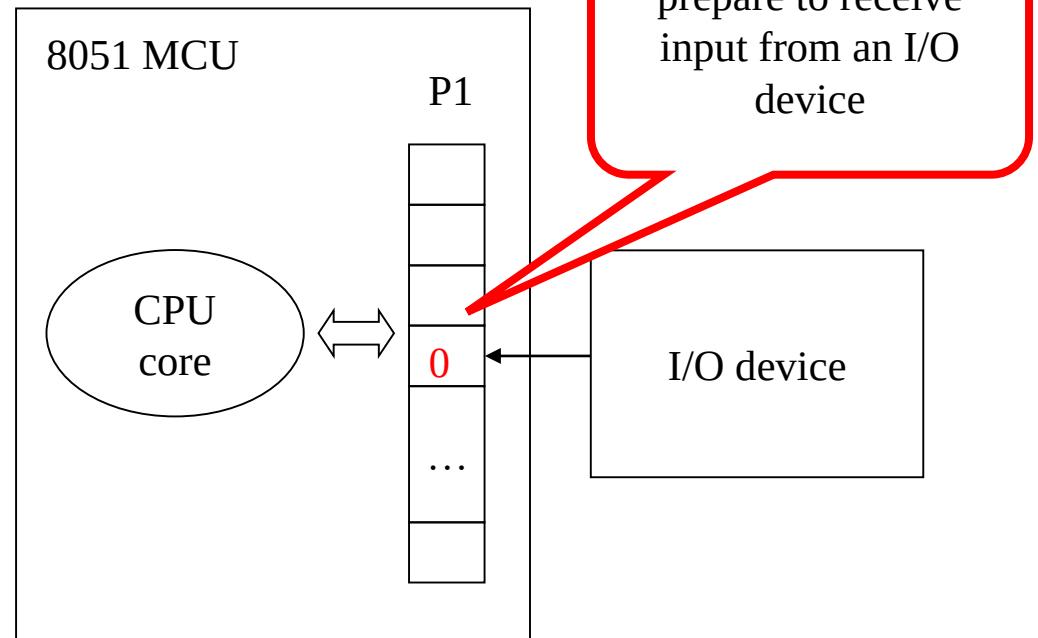
wait:

mov A, P1

JZ wait

exit:

//something after button pressed



Demo: wait for a button pressed

wait:

A = P1;

if (A==0) goto wait;

exit:

//something after button pressed

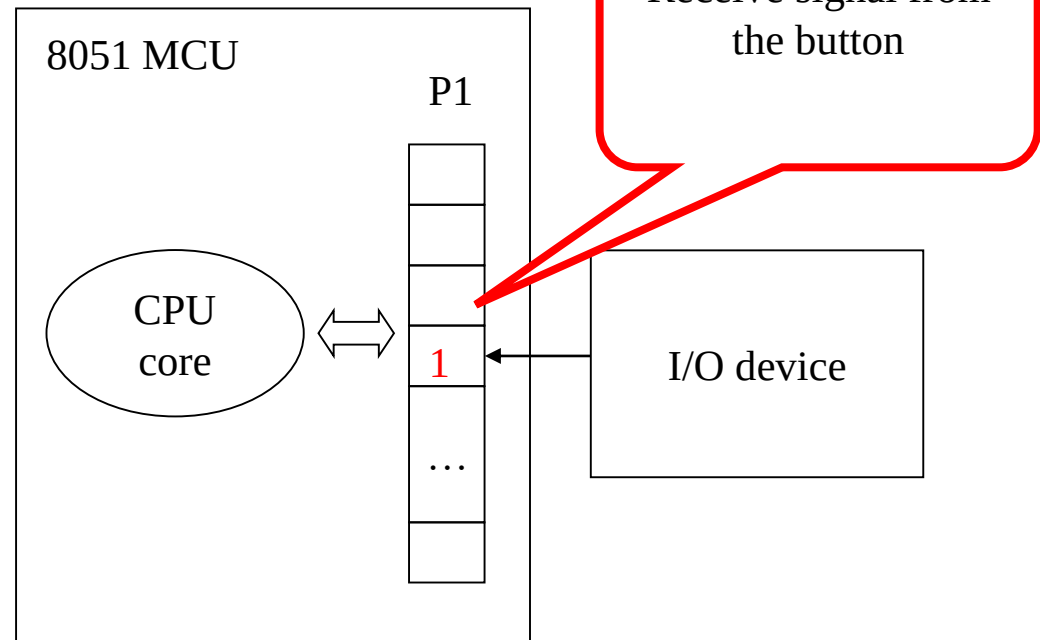
wait:

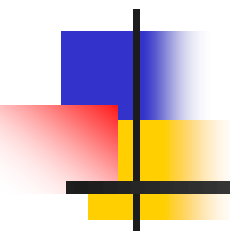
mov A, P1

JZ wait

exit:

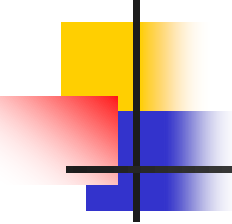
//something after button pressed





Example 2: make LED run

Show how to output signal



Demo: rotate the LED light

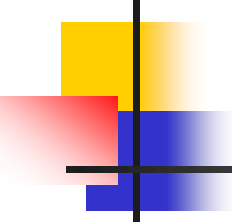
MAIN:

```
MOV A, #00000001B
MOV PSW, #00H
```

Loop:

```
MOV    P0, A
LCALL  Delay
RR      A
LJMP   Loop
```

```
                MOV    R0, #50
Delay:          MOV    R1, #40
Delay1:         MOV    R2, #249
Delay2:         DJNZ   R2, Delay2
                DJNZ   R1, Delay1
                DJNZ   R0, Delay
                RET
```



Demo: rotate the LED light

MAIN:

```
MOV A, #00000001B
MOV PSW, #00H
```

Loop:

```
MOV    P0, A
LCALL  Delay
RR      A
LJMP   Loop
```

control the LED through
content of A

```
                MOV    R0, #50
Delay:          MOV    R1, #40
Delay1:         MOV    R2, #249
Delay2:         DJNZ   R2, Delay2
                DJNZ   R1, Delay1
                DJNZ   R0, Delay
                RET
```

Demo: rotate the LED light

MAIN:

```
MOV A, #00000001B
MOV PSW, #00H
```

Loop:

```
MOV    P0, A
LCALL  Delay
RR      A
LJMP   Loop
```

- rotate right (RR) A

00000001

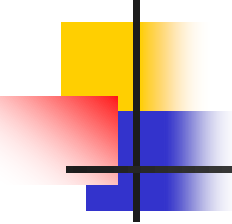


1000000



01000000

```
Delay:      MOV    R0, #50
Delay1:     MOV    R1, #40
Delay1:     MOV    R2, #249
Delay2:     DJNZ   R2, Delay2
            DJNZ   R1, Delay1
            DJNZ   R0, Delay
            RET
```



Demo: rotate the LED light

MAIN:

```
MOV A, #00000001B
MOV PSW, #00H
```

Loop:

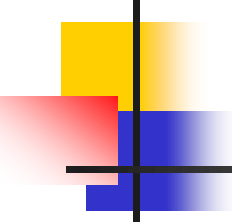
```
MOV    P0, A
LCALL  Delay
RR     A
LJMP   Loop
```

call a function at label
“Delay”

```
MOV    R0, #50
```

```
Delay:  MOV    R1, #40
Delay1: MOV    R2, #249
Delay2: DJNZ   R2, Delay2
        DJNZ   R1, Delay1
        DJNZ   R0, Delay
        RET
```

a nested loop to delay some
time

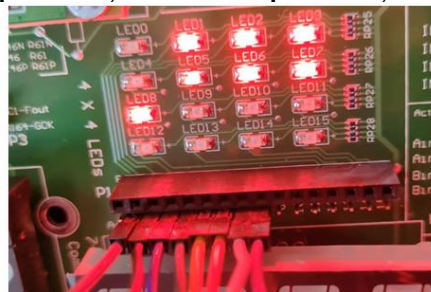


Demo Requirements

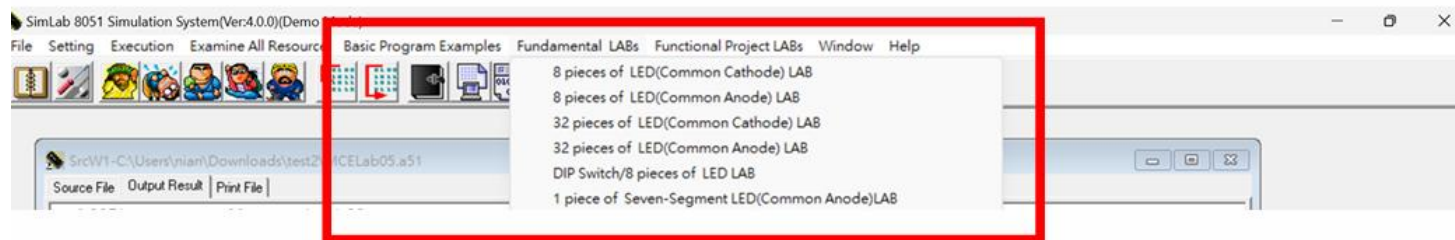
- Assembly only!
- Adjust the nested loop (in the previous slide) to let the LED light shift one step for EACH SECOND
- Bonus 1 (於結報10%加分)
 - 請將學號的數字部分加總後對100取餘數得到XY
 - 改成每 $(X+1)/2$ 秒移動一格
 - Y對5取餘數得到Z
 - 改成每次亮連續的Z+1顆LED燈
- Bonus 2 (於結報 5% or 10%加分)
 - 設計其他LED燈變換的樣態，助教會從兩種程度的加分擇一
 - 給分依據：這個樣態是不是很多人做一樣的?是否能明確解釋source code?

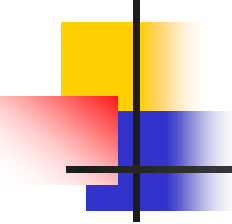
Hints

- 參考助教前次實驗給大家練習的軟體和硬體操作
- 請用杜邦線連接對應的pin腳
 - 板上對應的腳位 (P0.0 → pin 11; P0.1 → pin 12; P0.2 → 13; ...)



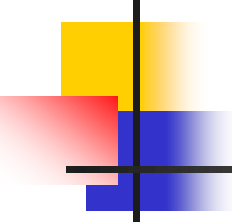
- SimLab裡有些參考範例可以協助大家快速上手此次實驗





Lab02 Study Report

- File name: Bxxxxxxx-MCE-Lab2-Study
- File type: PDF only
- The requirements of report
 - Summarize the content of this slide set
 - Provide your plan for this lab exercise
 - No more than one A4 page
 - Grading: 80 ± 15
- Deadline: 2026/10/7 23:00 (不收遲交)
- Upload to e-learning system



Lab02 Lab Exercise Report

- File name: Bxxxxxxx-MCE-Lab2-Result
- File type: PDF only
- The requirements of report
 - Summarize the problems and results you have in this exercise
 - Some screen shots or some code explanation can be provided
 - No more than two A4 pages
 - Grading: 80 ± 15
- Deadline: 2026/10/14 23:00 (不收遲交)
- Upload to e-learning system
- Bonus:
 - Read the Demo Requirements